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# Impact of Layer Norm on Memorization and Generalization in Transformers

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## Abstract

Layer Normalization (LayerNorm) is one of the fundamental components in transformers that stabilizes training and improves optimization. In recent times, Pre-LayerNorm transformers have become the preferred choice over Post-LayerNorm transformers due to their stable gradient flow. However, the impact of LayerNorm on learning and memorization across these architectures remains unclear. In this work, we investigate how LayerNorm influences memorization and learning for Pre- and Post-LayerNorm transformers. We identify that LayerNorm serves as a key factor for stable learning in Pre-LayerNorm transformers, while in Post-LayerNorm transformers, it impacts memorization. Our analysis reveals that eliminating LayerNorm parameters in Pre-LayerNorm models exacerbates memorization and destabilizes learning, while in Post-LayerNorm models, it effectively mitigates memorization by restoring genuine labels. We further precisely identify that early layers LayerNorm are the most critical over middle/later layers and their influence varies across Pre and Post LayerNorm models. We have validated it through 13 models across 6 Vision and Language datasets. These insights shed new light on the role of LayerNorm in shaping memorization and learning in transformers[2].

## 1 Introduction

Layer Normalization [Lei Ba et al., 2016] greatly contributes to stabilizing training and optimizing performance in deep learning models, especially in transformers. It works by normalizing the activations at each layer, ensuring a more consistent gradient flow during training. In recent years, transformers are primarily designed with two options by LayerNorm (LN) placements: *Pre-LN* and *Post-LN*. Post-LN transformer, introduced by Vaswani et al. [2017], applies normalization after the addition of the layer's output with the residual connection's output and has been showing competent performance in language modeling and machine translation. However, due to the issue of unstable gradient flow [Liu et al., 2020] in Post-LN models, Pre-LN transformers [Xiong et al., 2020] were introduced, where normalization is applied before self-attention and feed-forward layers. This configuration stabilized training and achieved faster convergence by improving the gradient flow, making it the preferred choice in modern architectures such as GPT, Llama, and Vision Transformers.

Even though transformers have demonstrated remarkable capabilities in learning rich representations from data, they exhibit a strong tendency to memorize some samples due to their complex nature, which is commonly known as *Label Memorization* [Feldman, 2020, Feldman and Zhang, 2020], where a model memorizes labels in training without learning the relevant patterns that generalize to unseen data, leading to overfitting. Recent studies have explored whether memorization can be localized to specific layers [Maini et al., 2023, Baldock et al., 2021] or components such as attention heads and the feed-forward network (FFN) [Haviv et al., 2023, Geva et al., 2023, Yu et al., 2023]. Apart from the attention heads and the feed-forward network (FFN), LayerNorm stands as a

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[2]Code available at: `https://github.com/JEKimLab/NeurIPS2025_LayernormMemorization`

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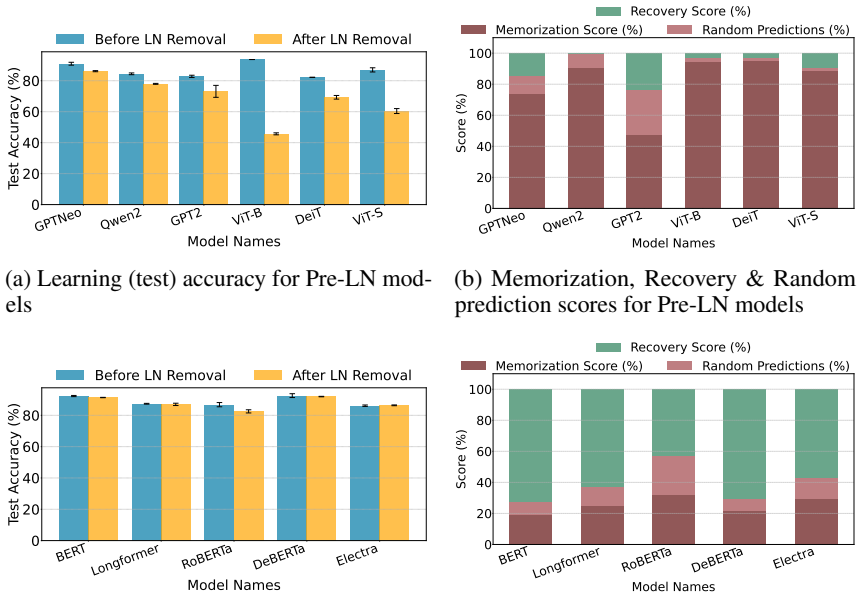


(a) Learning (test) accuracy for Pre-LN models

(b) Memorization, Recovery & Random prediction scores for Pre-LN models

(c) Learning (test) accuracy for Post-LN models

(d) Memorization, Recovery & Random prediction scores for Post-LN models

Figure 1: **Impact of LN layer on memorization and learning of Pre- and Post-LN models.** (a) shows a clear impact of LN in Pre-LN models, whereas (c) shows no impact of the removal of LN parameters in Post-LN models for learning. (b) exhibits that, without LN layers, the Pre-LN models struggle with high memorization and random predictions (red-color-family bars), while (d) exhibits that in Post-LN models, removing LN parameters recovers a significant portion of correct predictions (green bars).

pivotal component in the transformer architecture that further shapes its optimization dynamics and performance. Outlier neurons in LN layers have been shown to impair transformer performance and hinder quantization [Kovaleva et al., 2021, Puccetti et al., 2022, Bondarenko et al., 2023, He et al., 2024]. [Xu et al., 2019] hints that LN may contribute to increased overfitting in Pre-LN models.

In our paper, we identify that in Pre-LN transformers, LN is critical to *learning* and removal of its learnable parameters exacerbates overfitting and disrupts learning. On the contrary, in Post-LN transformers, LN plays a significant role in *memorization*, where by eliminating LN parameters, memorization is suppressed by recovering the true genuine labels without affecting the model's learning capability. We rigorously validate our claims through various models - BERT, Longformer, RoBERTa, DeBERTa, ELECTRA, DistilBERT, GPTNeo, GPT2, Qwen2, RoBERTa-PreLayerNorm, ViT-Base, ViT-Small, and DeiT, across both Vision and Language tasks. In summary, the core findings of our paper regarding the impact of LN on memorization and learning in transformers are as follows:

- **Learning Stability, Memorization Suppression & Label Recovery:** We identify that LN is crucial for learning in Pre-LN models, unlike Post-LN models. For Post-LN models, LN learnable parameters removal suppresses memorization and recovers genuine labels, whereas in Pre-LN models, LN removal exacerbates overfitting, with persistent memorization.

- **Early LNs are Critical:** We uncover that removal of LNs parameters in early layers is most impactful in mitigating memorization for Post-LN models, and destabilizing the learning in Pre-LN architectures.

- **Gradients Explain LN's Impact:** We explain the divergent impacts of LN in Pre- and Post-LN models by comparing learning and memorization gradients, which reveal why LN parameter removal causes learning disruption and memorization suppression in Pre- and Post-LN models, respectively.

## 2 Related Works

**Memorization & Learning:** Transformers excel at learning general, simple patterns [Arpit et al., 2017, Shah et al., 2020, Zhou and Wu, 2023], but also tend to memorize rare, mislabeled, or complex examples [Stephenson et al., 2021, Baldock et al., 2021, Agarwal et al., 2022]. Feldman and Zhang [2020], Feldman [2020] formally define label memorization, while Baldock et al. [2021] proposes prediction depth to capture example difficulty. Other works [Jiang et al., 2020, Ravikumar et al., 2024, Garg et al., 2023] associate high curvature and consistency with long-tailed or mislabeled samples. Beyond identifying memorization, several studies [Haviv et al., 2023, Geva et al., 2023, Dai et al., 2021] investigate how self-attention and feedforward layers contribute to factual recall across transformer layers. More recent work [Yin et al., 2023, Lad et al., 2024, Men et al., 2024, Li et al., 2024, Sun et al., 2025] highlights the limited effectiveness of deeper layers on learning in Pre-LN transformers. Despite these insights, the distinctive impact of LayerNorm in shaping memorization and learning across both Pre- and Post-LN architectures remains poorly understood.

**Understanding LayerNorm (LN) in Transformers:** In addition to self-attention and feedforward networks (FFNs), Layer Normalization (LN) plays a critical role in transformer models. Prior work [Brody et al., 2023, Wu et al., 2024] has demonstrated that LN is essential to the overall expressivity of transformers. Beyond its utility, LN has been found to contain outlier neurons [Kovaleva et al., 2021, Puccetti et al., 2022], whose removal severely degrades model performance. These outliers have also been shown to hinder the quantization of transformer models [Bondarenko et al., 2023, He et al., 2024]. Moreover, several studies [Xiong et al., 2020, Liu et al., 2020, Takase et al., 2022, Xie et al., 2023, Kim et al., 2025] have highlighted that Post-LN architectures can cause gradient instability during training, while Pre-LN configurations may lead to exploding gradients in early layers—prompting the development of techniques [Shleifer et al., 2021, Wang et al., 2022, Kumar et al., 2023, Qi et al., 2023, Jiang et al., 2023] to address them. Additionally, Xu et al. [2019] suggested that LN parameters may contribute to overfitting in Pre-LN models.

However, we provide a far more nuanced understanding of LN's role: in Pre-LN transformers, LN is essential for learning but not memorization, whereas in Post-LN models, LN is crucial for memorization but not learning. This distinction offers a novel contribution to understanding the function of LN in transformers for learning and memorization.

## 3 Prelimnaries

### 3.1 Understanding LayerNorm in Transformers and Defining Memorization & Learning

**LayerNorm Operation.** Let $x = (x_1, x_2, \ldots, x_d)$ be the input of size $d$ to the LayerNorm function $\text{LN}(x)$ which first normalizes the input $x$ as $N(x)$ using mean $\mu$ and standard deviation $\sigma$. Then it re-scales and re-centers $N(x)$ using the learnable parameters $w$ (weight) and $b$ (bias). The output of the LayerNorm layer is then given by:

$$\text{LN}(x) = w \odot N(x) + b, \quad \mu = \frac{1}{d}\sum_{i=1}^{d} x_i, \quad \sigma = \sqrt{\frac{1}{d}\sum_{i=1}^{d}(x_i - \mu)^2}, \quad N(x) = \frac{x - \mu}{\sigma}, \tag{1}$$

where $\odot$ denotes the dot product operation.

**Pre-LN & Post-LN Transformers.** In the Pre-LN Transformer, LayerNorm is applied before each sub-layer - Multi-Head Self Attention (MHSA) and Feed-Forward Network (FFN). On the other hand, in the Post-LN Transformer, LN is applied after the residual connection. We represent the key difference in the architectural design of the two configurations as follows:

$$\textbf{Pre-LN:} \quad \begin{aligned} x' &= x + \text{MHSA}(\text{LN}_1(x)) \\ y &= x' + \text{FFN}(\text{LN}_2(x')) \end{aligned} \qquad \textbf{Post-LN:} \quad \begin{aligned} x' &= \text{LN}_1(x + \text{MHSA}(x)) \\ y &= \text{LN}_2(x' + \text{FFN}(x')) \end{aligned} \tag{2}$$

**Understanding Learning and Label Memorization (LM).** Deep neural network models, like transformers, learn meaningful relationships between features and labels during training and generalize the learned representations to unseen test data - the phenomenon is well understood as ***learning/generalization***. At the same time, these models also have the tendency to memorize training data points which are complex in nature, commonly known as ***label memorization (LM)*** [Feldman, 2020, Feldman and Zhang, 2020], where the model just memorizes the labels during training without